

ARIANNA FERNANDEZ

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Education

University of Massachusetts Boston

GPA: 4.00/4.00

B.S. Computer Science | Minors: Economics & Mathematics

Dean's List | Alpha Lambda Delta Honor Society

Professional Experience

WebAI Gen | Software Engineering | Hybrid

June 2026 - September 2026

- Developed **8+** responsive full-stack web applications using CSS, SQL, HTML, React, Next.js, TypeScript, and REST APIs for startups across multiple industries.
- Improved page performance and SEO, increasing Lighthouse performance scores by **25–35%**.
- Collaborated with 4+ engineers across 15+ product releases, contributing to testing, debugging, deployment, and ML dataset preparation.

Supplemental Instruction Leader | Data Structures and Algorithms

September 2025- Ongoing

- Led weekly review sessions for 8+ undergraduate students, improving exam scores by 20%.
- Developed **20+** review worksheets and coding exercises in Java covering linked lists, arrays, queues, stacks, recursion, trees, sets, and hash tables.
- Held **5-8** office hours/tutorial sessions per week, supporting one-on-one debugging and problem solving.

Extern | AI/ML Document Intelligence Externship

March 2026 - June 2026

- Developed OCR and NLP pipelines in Python, processing **2,000+** mortgage documents into structured datasets.
- Prototyped RAG systems with LlamaIndex and evaluated LLMs for document understanding.
- Achieved **95%+** field extraction accuracy through preprocessing and validation.

Research Experience

Machine Learning Fairness on Credit Approval Models | Fellowship

January 2026-May 2026

- Designed and developed an interpretable machine learning model for consumer credit approval, improving approval rates by 29% while maintaining predictive performance.
- Investigated algorithmic bias, proxy variables, and socioeconomic factors affecting predictive outcomes.
- Applied statistical analysis and data processing for model evaluation.
- Evaluated **5+** fairness metrics (Equal Opportunity, Demographic Parity, Disparate Impact, etc.).

Research Assistant | IEEE ETHICS Conference (First Author)

October 2024-June 2025

- Conducted mixed-methods research using surveys, focus groups, and thematic analysis.
- Analyzed qualitative and quantitative datasets to identify trends influencing student persistence and academic outcomes of BIWOC women in Computer Science.
- Synthesized findings into a [first-author peer-reviewed IEEE publication presented at IEEE ETHICS](#).

ACTIVITIES & LEADERSHIP

Student Multicultural Affairs Community Leader & GROW Program Coordinator:

- Coordinated leadership programs, event logistics, communications, and operations supporting campus-wide initiatives

SKILLS

Languages: TypeScript, Python, JavaScript, Java, SQL, C, C++, HTML/CSS

Frameworks & Libraries: React, Next.js, LlamaIndex, REST APIs

Tools & Platforms: Git, GitHub, Power BI

AWARDS & HONORS

- World Robot Olympiads (Future Innovators): Ranked 2nd in the Americas and 1st in Latin America
- Paul English Applied Artificial Intelligence Institute: Scholarship and Hackathon Outstanding Initiative Award.